A clearer maintenance report

Design, implementation and evaluation of a two-state JavaScript application

A tenant can hear a tap dripping, but the property manager cannot see which fixture is affected. Haven asks for the room, the observed problem and when it began. The tenant then supplies a category and urgency so the summary can be read at a glance.

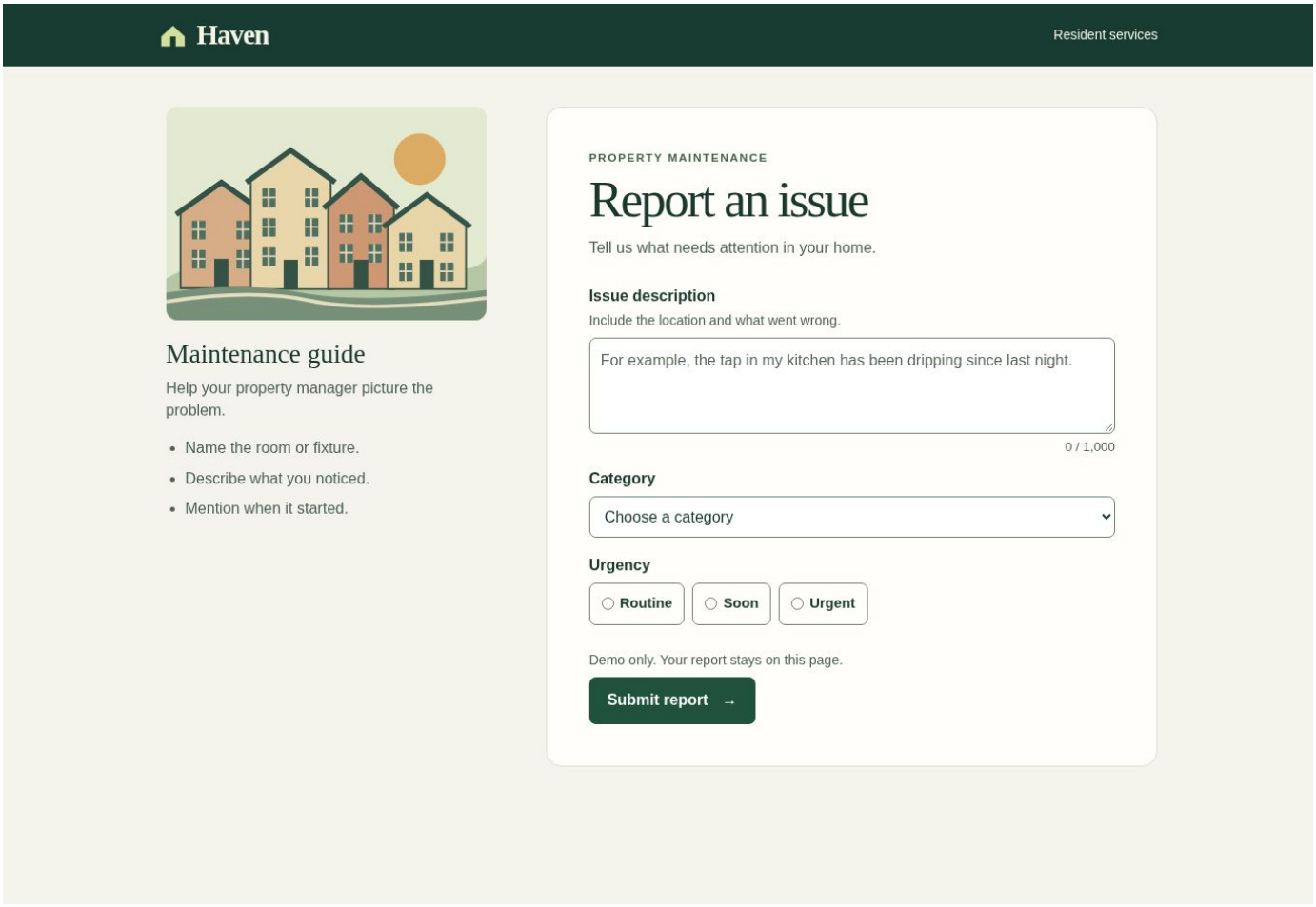


FIG. 1 Desktop entry screen captured in Chrome.

What happens on submit?

The browser checks the required fields. JavaScript reads the form and displays the description, category and urgency together. All values live in the current page. Refreshing or starting another report clears them. The demo sends no email and creates no work order.

From desktop to phone

One form, two layouts and the same reading order within the task

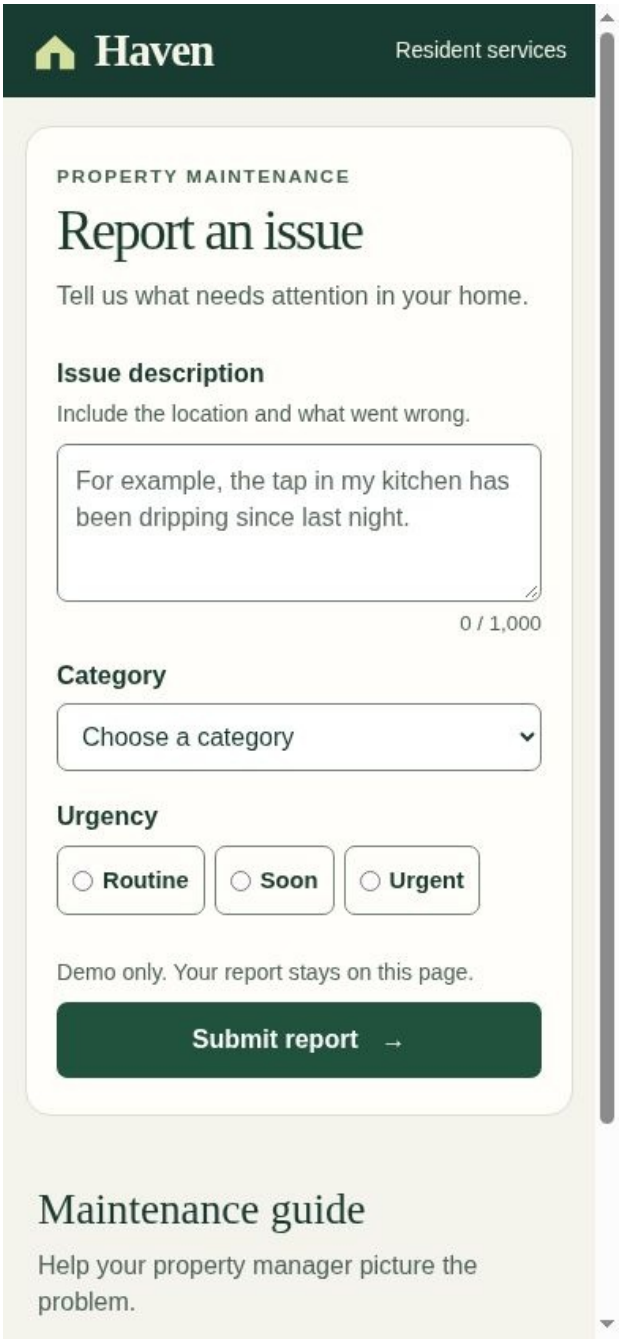


FIG. 2 Phone-width entry screen.

Tested layouts

The desktop capture is 1363 × 936. The 390 × 844 capture at left is the actual app rendered in a phone-sized browser frame. Both show the complete form within the viewport width.

On desktop, the guide sits beside the form. At 720 pixels and below, the form comes first and the guide moves underneath. The streetscape disappears so it cannot push the first field down.

On narrow screens the button spans the form. Urgency options keep a minimum height of 44 pixels and can wrap on smaller widths.

The phone capture shows all three choices and the submit button together. A vertical scroll reaches the rest of the guide. This sample gave no reason to reduce the text size or tighten the controls further.

Verify the interaction

Keyboard walkthrough, contrast checks and one remaining test



FIG. 3 Actual confirmation. The outline marks keyboard focus.

Test case

A kitchen tap drips even when its handles are closed. The expected summary preserves the description and shows Plumbing and Soon.

Keyboard evidence

In Chrome, Tab reached the description and then the category. Typing P selected Plumbing. Tab entered the radio group and Right selected Soon. Tab and Enter submitted the form. The accessibility tree identified the focused heading as “Report confirmed.” Tab and Enter on “Report another issue” cleared the form and returned focus to the description.

Labels and contrast

Visible labels name the textarea and category selector. A fieldset and legend identify the urgency group. Calculated contrast was 6.56:1 for helper text (#4e6157) on the panel (#fffefa) and 8.98:1 for white text on the green button (#20523e).

The browser tree confirms names, roles and focus. A screen-reader listening test remains open, particularly for native validation messages that differ across browsers.

Human and AI collaboration

Division of work, two design decisions and the student review

The student selected the maintenance scenario, fixed the scope and requested JavaScript and VS Code. ChatGPT/Codex drafted the interface and code, operated Chrome and assembled the evidence. The browser supplied observable results for claims that code generation alone could only predict.

Task	AI role	Human decision	Verification method
Define scope	Translate the selected flow into two states.	Student chose the scenario and excluded delivery services.	Compare the UI to the assignment brief.
Build and test	Write HTML, CSS and JS. Run a keyboard example.	Student to review field choices and demonstrate locally.	Compare entered values with the receipt.
Publish and reflect	Prepare boards, source files and evidence.	Student controls public sharing and personal reflection.	Open all three links while signed out.

1. A confirmation the prototype can support

The brief says “submit,” but the app has no delivery service. The implementation therefore says “Demo only” before submission and shows a local receipt afterward. The keyboard test confirmed the receipt reproduced the tenant’s choices without promising that a manager had received them.

2. Validate the text behind the label

A required textarea accepts spaces as input. The implementation adds a trimmed-value check with setCustomValidity and clears that error when the description changes. Reproduce the edge case by typing spaces, choosing a category and urgency, and submitting. Replace the spaces with a real description and submit again.

Student verification before submission

Run both examples yourself. In your own words, explain why the report disappears on refresh, which AI-assisted decision you accept or would change, and what you personally verified. Add your Figma, public repository and deployed site URLs. Sharing and final sign-off remain with the student.